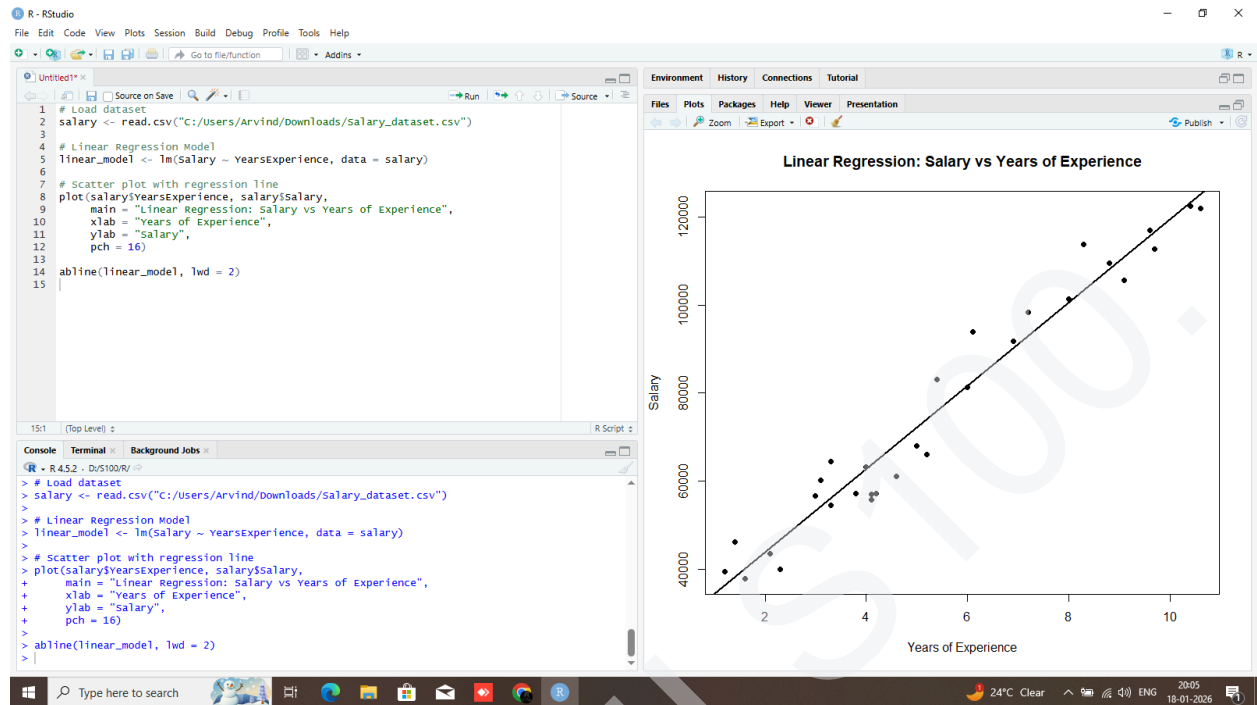


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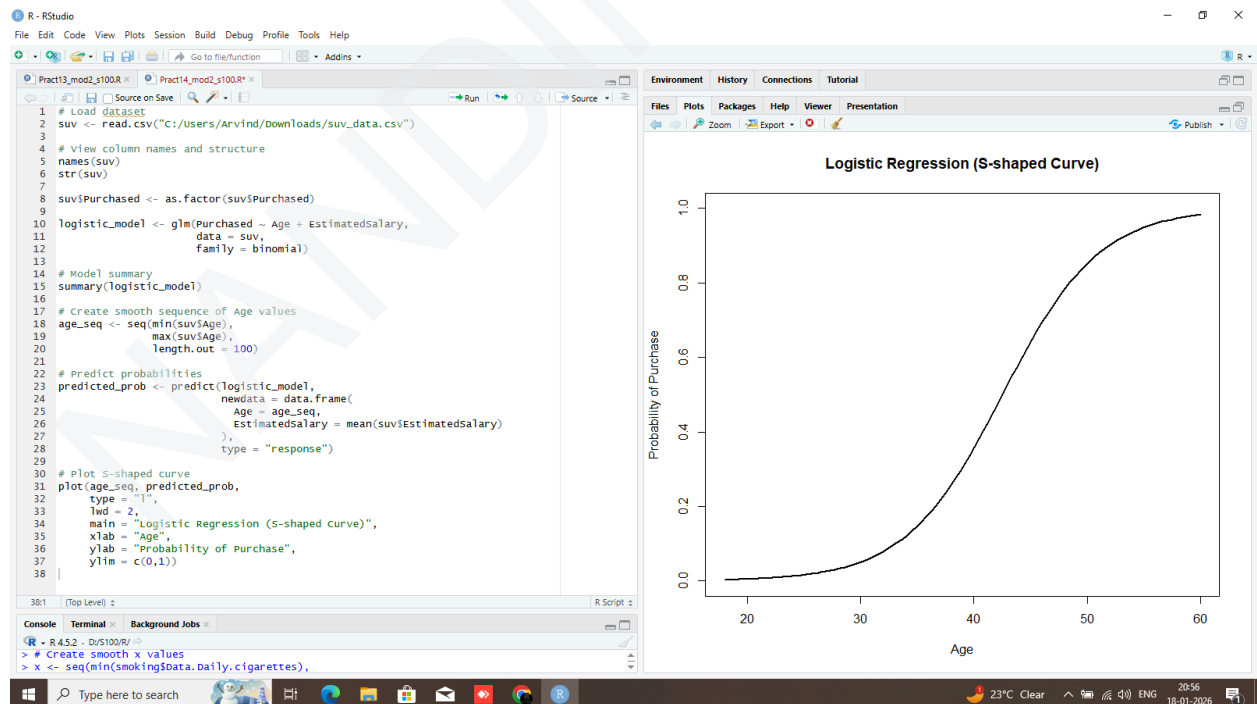
Practical no : 13

Aim :- Performing linear regression analysis using `lm()` (R).



Practical no : 14

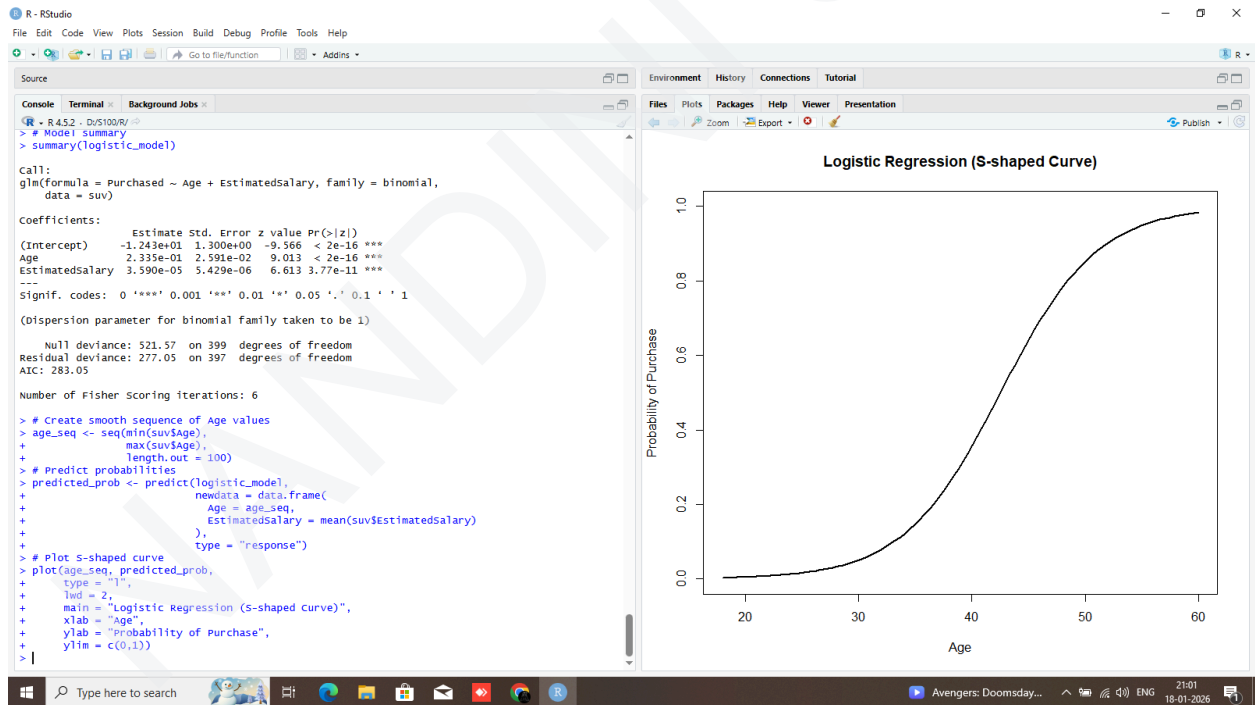
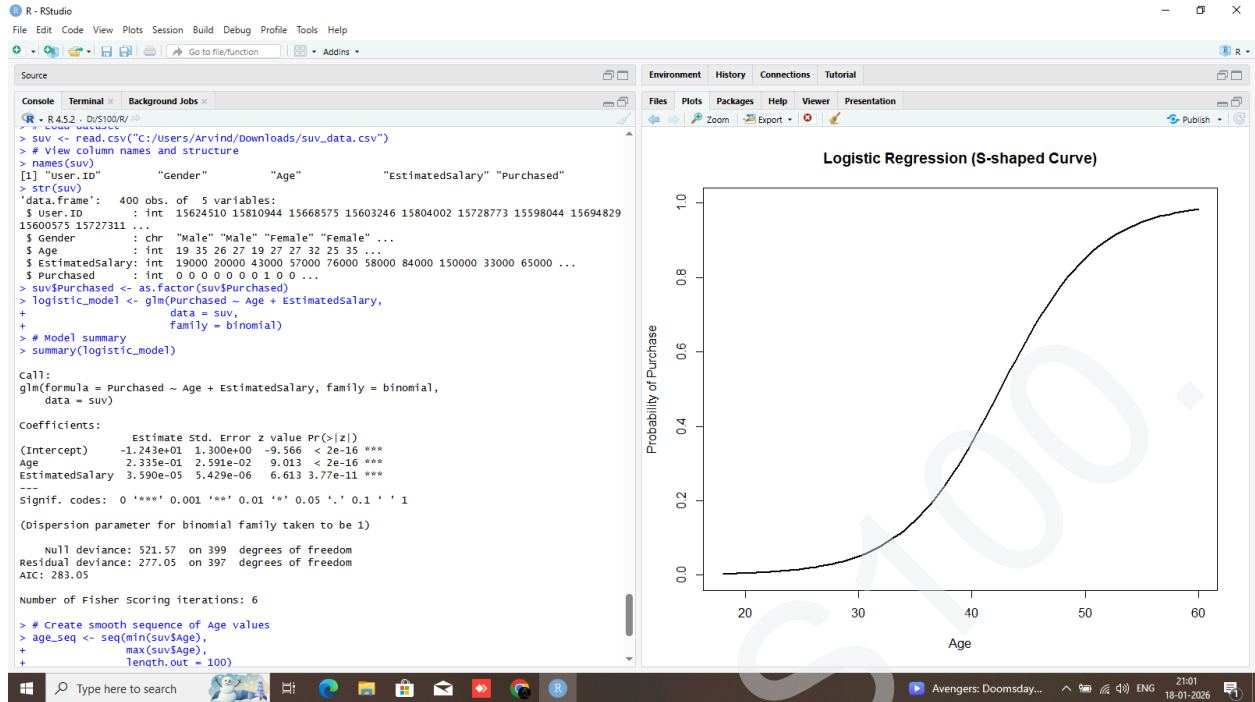
Aim :- Performing logistic regression using `glm()` (R).



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DATA ANALYSIS WITH SAS/SPSS/R

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Practical no : 15

Aim :- Exporting results into external files (Excel, CSV, PDF) using write.csv() and writexl ®

Code :-

```
install.packages("writexl")
library(writexl)

#pract13
write.csv(salary, "C:/Users/itlab/Downloads/Salary_dataset_export.csv", row.names = FALSE)

# Add predictions to dataset
salary$Predicted_Salary <- predict(linear_model, salary)

# Export to CSV
write.csv(salary, "C:/Users/itlab/Downloads/Salary_with_predictions.csv", row.names = FALSE)

write_xlsx(salary, "C:/Users/itlab/Downloads/Salary_with_predictions.xlsx")

#PDF device
pdf("C:/Users/itlab/Downloads/Salary_Regression_Plot.pdf")

# Plot inside PDF
plot(salary$YearsExperience, salary$Salary,
     main = "Linear Regression: Salary vs Years of Experience",
     xlab = "Years of Experience",
     ylab = "Salary",
     pch = 16)
abline(linear_model, lwd = 2)

# Close PDF device
dev.off()

#pract14
# Add predicted probabilities for the existing dataset
suv$Predicted_Prob <- predict(logistic_model, suv, type = "response")

# Export to CSV
write.csv(suv, "C:/Users/itlab/Downloads/SUV_with_predictions.csv", row.names = FALSE)

pred_df <- data.frame(Age = age_seq,
                      EstimatedSalary = mean(suv$EstimatedSalary),
                      Predicted_Prob = predicted_prob)

write.csv(pred_df, "C:/Users/itlab/Downloads/Predicted_Probabilities.csv", row.names = FALSE)
write_xlsx(pred_df, "C:/Users/itlab/Downloads/Predicted_Probabilities.xlsx")
```

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```
#Open PDF device
```

```
pdf("C:/Users/itlab/Downloads/Logistic_Regression_Plot.pdf")
```

```
# Plot the S-shaped probability curve
```

```
plot(age_seq, predicted_prob,
```

```
  type = "l",
```

```
  lwd = 2,
```

```
  main = "Logistic Regression (S-shaped Curve)",
```

```
  xlab = "Age",
```

```
  ylab = "Probability of Purchase",
```

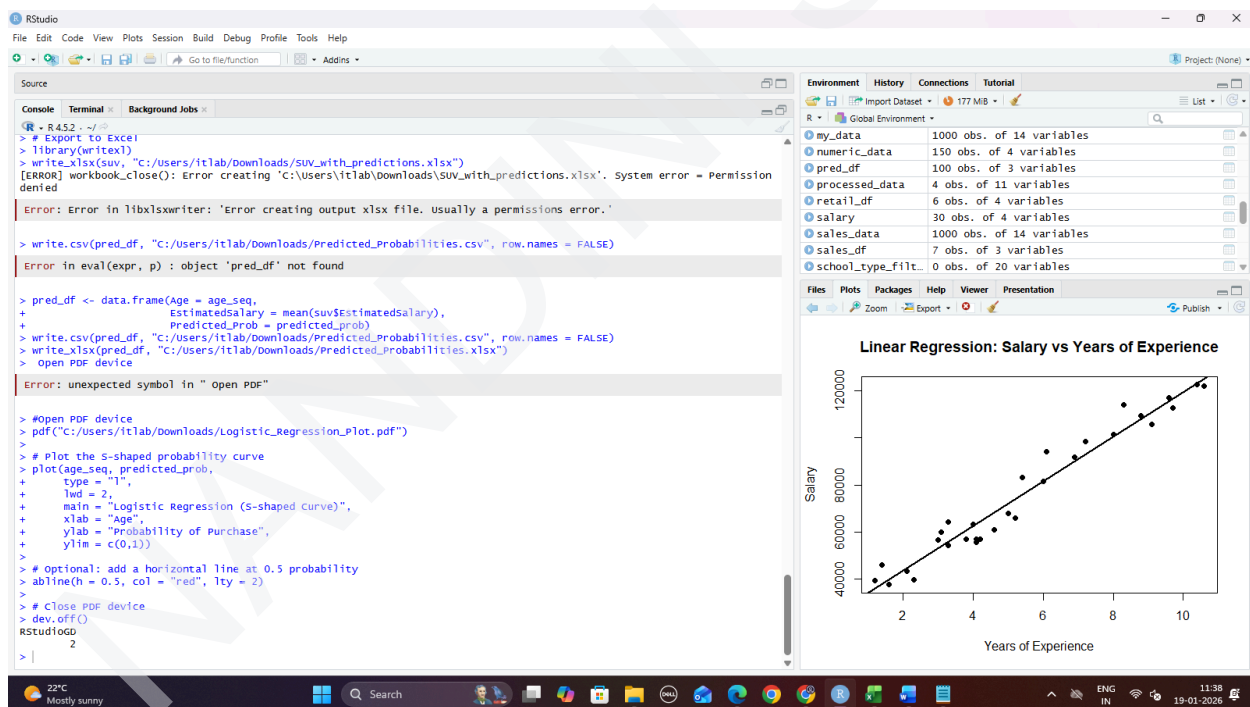
```
  ylim = c(0,1))
```

```
# Optional: add a horizontal line at 0.5 probability
```

```
abline(h = 0.5, col = "red", lty = 2)
```

```
# Close PDF device
```

```
dev.off()
```



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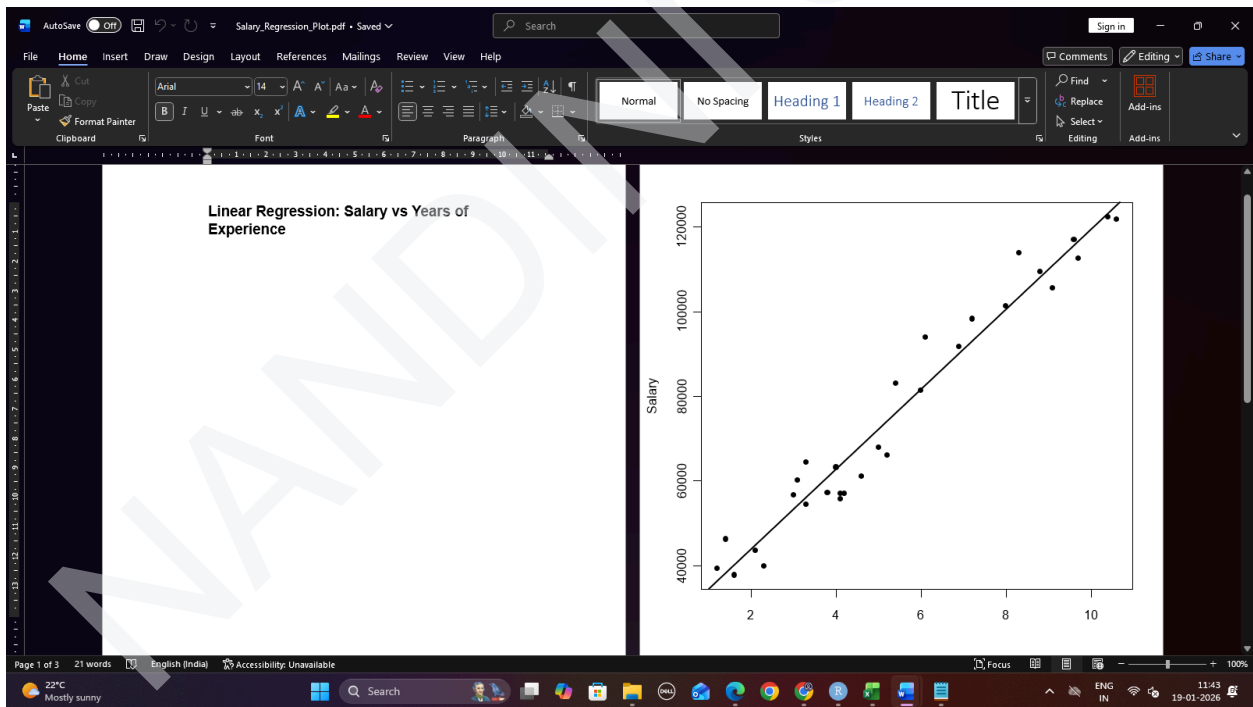
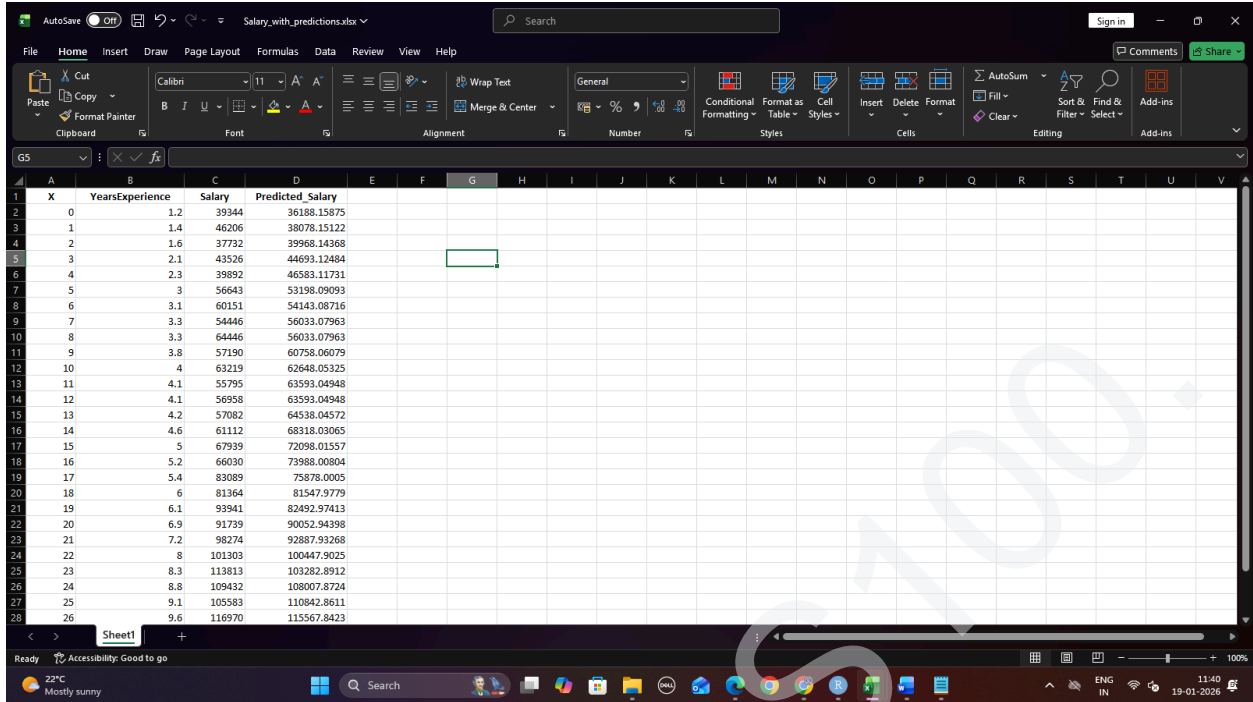
DATA ANALYSIS WITH SAS/SPSS/R

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	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	X	YearsExperience	Salary																			
2		0	1.2	39344																		
3		1	1.4	46206																		
4		2	1.6	37732																		
5		3	2.1	43526																		
6		4	2.3	39892																		
7		5	3	56643																		
8		6	3.1	60151																		
9		7	3.3	54446																		
10		8	3.3	64446																		
11		9	3.8	57190																		
12		10	4	63219																		
13		11	4.1	55795																		
14		12	4.1	56958																		
15		13	4.2	57082																		
16		14	4.6	61112																		
17		15	5	67939																		
18		16	5.2	66030																		
19		17	5.4	83089																		
20		18	6	81364																		
21		19	6.1	93941																		
22		20	6.9	91739																		
23		21	7.2	98274																		
24		22	8	101303																		
25		23	8.3	113813																		
26		24	8.8	109432																		
27		25	9.1	105583																		
28		26	9.6	116970																		

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	X	YearsExperience	Salary	Predicted_Salary																		
2		0	1.2	39344	36188.15875																	
3		1	1.4	46206	38078.15122																	
4		2	1.6	37732	39968.14368																	
5		3	2.1	43526	44693.12484																	
6		4	2.3	39892	46583.11731																	
7		5	3	56643	53198.09093																	
8		6	3.1	60151	54143.08716																	
9		7	3.3	54446	56033.07963																	
10		8	3.3	64446	56033.07963																	
11		9	3.8	57190	60758.06079																	
12		10	4	63219	62648.05325																	
13		11	4.1	55795	63593.04948																	
14		12	4.1	56958	63593.04948																	
15		13	4.2	57082	64538.04572																	
16		14	4.6	61112	68318.03065																	
17		15	5	67939	72096.01557																	
18		16	5.2	66030	73988.00804																	
19		17	5.4	83089	75878.00005																	
20		18	6	81364	81547.0779																	
21		19	6.1	93941	83492.97413																	
22		20	6.9	91739	90052.94398																	
23		21	7.2	98274	92887.93268																	
24		22	8	101303	100447.9025																	
25		23	8.3	113813	103282.8912																	
26		24	8.8	109432	108007.8724																	
27		25	9.1	105583	110842.8611																	
28		26	9.6	116970	115567.8423																	

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DATA ANALYSIS WITH SAS/SPSS/R

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Practical 14

User.ID	Gender	Age	EstimatedSalary	Purchased	Predicted_Prob
15624510	Male	19	19000	0	0.000664659
15810944	Male	35	20000	0	0.028090293
15668575	Female	26	43000	0	0.008006028
15603246	Female	27	57000	0	0.016570537
15804002	Male	19	76000	0	0.005121275
15728773	Male	27	58000	0	0.017165845
15598044	Female	27	84000	0	0.042529606
15694829	Female	32	150000	1	0.604156606
15600575	Male	25	33000	0	0.004442785
15727311	Female	35	65000	0	0.126939237
15570769	Female	26	80000	0	0.029564015
15606274	Female	26	52000	0	0.011025979
15746139	Male	20	86000	0	0.009223645
15704987	Male	32	18000	0	0.013175814
15628972	Male	18	82000	0	0.005029957
15697686	Male	29	80000	0	0.057827757
15733883	Male	47	25000	1	0.362978257
15617482	Male	45	26000	1	0.270213665
15704583	Male	46	28000	1	0.334423043
15621083	Female	48	29000	1	0.453790327
15649487	Male	45	22000	1	0.242845378
15736760	Female	47	49000	1	0.574237565
15714658	Male	48	41000	1	0.561054699
15599081	Female	45	22000	1	0.242845378
15705113	Male	46	23000	1	0.295722243
15631159	Male	47	20000	1	0.322574483
15792818	Male	49	28000	1	0.503055173

User.ID	Gender	Age	EstimatedSalary	Purchased	Predicted_Prob
15624510	Male	19	19000	0	0.000664659
15810944	Male	35	20000	0	0.028090293
15668575	Female	26	43000	0	0.008006028
15603246	Female	27	57000	0	0.016570537
15804002	Male	19	76000	0	0.005121275
15728773	Male	27	58000	0	0.017165845
15598044	Female	27	84000	0	0.042529606
15694829	Female	32	150000	1	0.604156606
15600575	Male	25	33000	0	0.004442785
15727311	Female	35	65000	0	0.126939237
15570769	Female	26	80000	0	0.029564015
15606274	Female	26	52000	0	0.011025979
15746139	Male	20	86000	0	0.009223645
15704987	Male	32	18000	0	0.013175814
15628972	Male	18	82000	0	0.005029957
15697686	Male	29	80000	0	0.057827757
15733883	Male	47	25000	1	0.362978257
15617482	Male	45	26000	1	0.270213665
15704583	Male	46	28000	1	0.334423043
15621083	Female	48	29000	1	0.453790327
15649487	Male	45	22000	1	0.242845378
15736760	Female	47	49000	1	0.574237565
15714658	Male	48	41000	1	0.561054699
15599081	Female	45	22000	1	0.242845378
15705113	Male	46	23000	1	0.295722243
15631159	Male	47	20000	1	0.322574483
15792818	Male	49	28000	1	0.503055173

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DATA ANALYSIS WITH SAS/SPSS/R

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Age	EstimatedSalary	Predicted_Prob
18	69742.5	0.003245106
18.42424242	69742.5	0.003581803
18.84848485	69742.5	0.003953295
19.27272727	69742.5	0.004363149
19.6969697	69742.5	0.004815288
20.12121212	69742.5	0.005314031
20.54545455	69742.5	0.005864127
20.96969697	69742.5	0.006470798
21.39393939	69742.5	0.00713978
21.81818182	69742.5	0.007877376
22.24242424	69742.5	0.008690506
22.66666667	69742.5	0.009586757
23.09090909	69742.5	0.010574454
23.51515152	69742.5	0.011662712
23.93939394	69742.5	0.012861511
24.36363636	69742.5	0.014181764
24.78787879	69742.5	0.015635398
25.21212121	69742.5	0.017235424
25.63636364	69742.5	0.018996027
26.06060606	69742.5	0.020932645
26.48484848	69742.5	0.023062057
26.90909091	69742.5	0.025402467
27.33333333	69742.5	0.027973589
27.75757576	69742.5	0.030796725
28.18181818	69742.5	0.03389484
28.60606061	69742.5	0.03729263
29.03030303	69742.5	0.041016571

Age	EstimatedSalary	Predicted_Prob
18	69742.5	0.003245106
18.42424242	69742.5	0.003581803
18.84848485	69742.5	0.003953295
19.27272727	69742.5	0.004363149
19.6969697	69742.5	0.004815288
20.12121212	69742.5	0.005314031
20.54545455	69742.5	0.005864127
20.96969697	69742.5	0.006470798
21.39393939	69742.5	0.00713978
21.81818182	69742.5	0.007877376
22.24242424	69742.5	0.008690506
22.66666667	69742.5	0.009586757
23.09090909	69742.5	0.010574454
23.51515152	69742.5	0.011662712
23.93939394	69742.5	0.012861511
24.36363636	69742.5	0.014181764
24.78787879	69742.5	0.015635398
25.21212121	69742.5	0.017235424
25.63636364	69742.5	0.018996027
26.06060606	69742.5	0.020932645
26.48484848	69742.5	0.023062057
26.90909091	69742.5	0.025402467
27.33333333	69742.5	0.027973589
27.75757576	69742.5	0.030796725
28.18181818	69742.5	0.03389484
28.60606061	69742.5	0.03729263
29.03030303	69742.5	0.041016571

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```
Call:
glm(formula = Purchased ~ Age + EstimatedSalary, family = binomial,
     data = suv)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.243e+01  1.300e+00 -9.566 < 2e-16 ***
Age          2.335e-01  2.591e-02  9.013 < 2e-16 ***
EstimatedSalary 3.590e-05  5.429e-06  6.613 3.77e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 521.57  on 399  degrees of freedom
Residual deviance: 277.05  on 397  degrees of freedom
AIC: 283.05

Number of Fisher Scoring iterations: 6
```

